STUDENTS' TECHNICAL COUNCIL

Tournafest

14-16 APRIL 2023

TECHSHILA

INTER BHAWAN TECH MEET

Software Development

Tournafest's Remote PC Controller Application Challenge

The objective is to create a Remote PC Controller Application that permits individuals to remotely supervise and handle the activities occurring on their personal computers using their smartphones. The application must have a user-friendly interface that enables users to terminate processes, power off, or put their PCs in hibernation mode.

Problem Statement

Meet Ayush, A final-year student at IIT Roorkee who is very busy with his responsibilities. He must complete his BTP and often keeps his laptop on, training his ML model while attending calls/meetings for hours. He wishes to be able to turn off his laptop using his phone so that he doesn't need to return to his room too many times.

You, a software developer, promise to help him by building an application to help him achieve this. Ayush also wants the application to be super easy to use, so you need to build a great UI and a simple UX. Ayush asks you to build an application that allows him to terminate processes, power off, or put his PC in hibernation mode.

Task

Your task is to build an application that can be accessed using your mobile device and can control your PC.

A basic architecture would include three units:

1. **A frontend (to manage your PC)**

The frontend must include a simple UI containing the control options for the PC (shut-down / hibernate / terminate-process / etc.). There must be a way to show the current status of each control for the PC. A login system is needed to gain control access to your PC. A simple login system could include using an access code generated on the PC that is entered on the frontend to access the PC. However, you can use a different approach for the same (login with username-password / Google / etc.) [brownie points].

2. **A backend (to handle routing and business logic)**

The backend will have routes for login/signup, access code generation, and PC control. Include all the business logic, APIs, middleware, etc., here.

3. **A desktop application (to gain access to the PC)**

This application will be installed on the PC and will have access to the PC's processes. It will be connected to the backend. A simple frontend for this application is essential, as the access code will be shown here. We recommend dockerizing this application or running it on a VM so that it can be tested quickly. This application could be a cross-platform application [brownie points].

Apart from shut-down / hibernate / terminate-process features, we encourage participants to ideate and add more features [brownie points] relevant to this PS.

Participation

Participation details related to the challenge:

1. The maximum team size is 8 members
2. Contact your bhawan's contingent leader to participate

Evaluation

The points distribution is as follows:

1. Total Event Points: 100
2. Ease of setup, Ease of use, and Number of working features: 50%
3. [Brownie points] Extra features, Ability to control and monitor multiple PCs at the same time, Better login system, Cross-platform compatibility (macOS, Linux & Windows): 20%
4. Presentation weightage: 30%

Submission

You need to submit the following:

1. GitHub repos containing the codebase with a Readme file that contains clear and concise setup instructions.
2. A short report containing your approach and choice of the tech stack.

The submission details are as follows:

- 14th April: Code and Report Submission
- 15th April: Interaction with the company

Note: Top Submissions will get a chance for an internship interview!

Resources

Following is a list of quick resources to help you get started, but researching is an integral part of the development experience and hence the list is not exhaustive. You are always welcome to ask for help on Discord 🙂

- https://ourcodeworld.com/articles/read/411/how-to-shutdown-and-reboot-linux-with-node-js
- https://www.geeksforgeeks.org/python-script-to-shutdown-computer/
- https://docs.python.org/3/library/os.html
- https://nodejs.org/api/os.html
- https://expressjs.com/
- https://www.docker.com/
- https://www.electronjs.org/
- https://react.dev/